DIEFEN: Differential Information-Enhanced Feature Exchange Network for Hyperspectral Change Detection

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***Abstract*— Hyperspectral image (HSI) change detection (CD) has gained significant attention in the field of remote sensing. Current CD methods typically extract features based on spatial or spectral correlations between bitemporal HSIs, which often overlook the difference information, leading to a decrease in CD accuracy. Furthermore, these algorithms do not fully consider the alignment of features between images across different channel and spatial dimensions. To tackle these issues, we propose a novel approach called the differential information-enhanced feature exchange network (DIEFEN) for HSI CD, which leverages the difference information between images and enhances the alignment of bitemporal image features to improve CD accuracy. Specifically, an enhanced differential multihead attention (EDMA) module is proposed to utilize difference information to guide the feature aggregation of bitemporal images, effectively highlighting changing pixels and suppressing unchanging pixels. A feature focus and long-range attention (FFLA) module is designed to extract local and global features, and a channel-spatial interaction (CSI) module is constructed to align features and mitigate the impact of noise. Experimental results on three HSI CD datasets demonstrate that the proposed DIEFEN method outperforms several state-of-the-art methods. The source code of the proposed DIEFEN is released at https://github.com/creativeXin/DIEFEN.**

***Index Terms*— Change detection (CD), feature exchange, hyperspectral image (HSI), multihead attention.**

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I. Introduction

CHANGE detection (CD) aims to accurately identify changes in the Earth surface or target by comparing geospatial data captured at different times. It is a crucial technology for monitoring Earth surface changes and has widely applied in environmental surveillance, urban planning, disaster response, agricultural monitoring, and forest conservation [1], [2], [3], [4], [5], [6]. Hyperspectral imaging technology can capture subtle differences in objects or surface materials across a wide spectral range, significantly enhancing the precision and sensitivity of CD [7], [8], [9].

Over the past few decades, many traditional methods have been proposed for hyperspectral image (HSI) CD. In general, these methods are divided into four categories: algebra-based methods, transformation-based methods, classification-based methods, and other advanced methods [10]. Early algebra-based methods include image differencing, image rationing, and vegetation index differencing. Change vector analysis (CVA) is a classical algebra-based method that can identify the binary change information through spectral vector subtraction and threshold calculation [11]. Transformation-based methods usually transform an image into a specific space, highlighting the changing information while suppressing the unchanging information. Deng et al. [12] introduced the principal component analysis (PCA) to enhance change information from stacked multisensor data after geometric correction and radiometric normalization. Ortiz-Rivera et al. [13] proposed temporal PCA (TPCA) for CD. Other scholars have also proposed subspace-based CD [14] and iteratively reweighted multivariate alteration detection (IR-MAD) [15] to perform CD. Classification-based algorithms include postclassification and direct classification methods [16], [17], [18]. Bovolo et al. [17] proposed a semisupervised support vector machine (SVM) to directly identify changes in multitemporal remote sensing images. Lv et al. [19] used band selection and CVA to obtain change magnitude image. In addition, there are many other methods for CD, such as unmixing-based methods [20] and tensor-based decomposition methods [21]. However, traditional methods may have limited feature extraction capability for CD.

In recent years, convolutional neural network (CNN) has gained significant attention in CD task due to its strong

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feature extraction capabilities [22], [23], [25]. For example, Seydi et al. [23] combined 1-D, 2-D, and 3-D-dilated convolutions to optimize feature extraction. Saha et al. [24] used the pretrained CNN for semantic segmentation of a single image to obtain deep features. Li et al. [26] proposed a multiscale fully CNN to extract detailed features of ground objects and enhance the effectiveness of detecting subtle changes. Du et al. [27] utilized two symmetric deep networks to project input data and highlight the transformed features' changed components. Wang et al. [28] proposed a 2-D convolution neural network to extract fusion features of HSI and abundance maps to learn different features and enhance generalization ability. Lin et al. [29] proposed a bilinear CNN to capture feature representation and exploit the relations between the two input images. Wu et al. [30] used the spatial, spectral, and temporal attention module to fuse the multidomain features. Ou et al. [31] utilized the CNN to extract spatial and spectral features of HSIs. Luo et al. [32] used CNN to extract multiscale information from HSIs and achieved better results for HSI CD. Mou et al. [33] combined a CNN and a recurrent neural network to generate both spectral and spatial feature representations, and to analyze the temporal dependence of bitemporal images. Although CNNs have been proven effective in CD, they have inherent limitations in capturing spatial relationships over long-term dependence in images due to their local receptive fields.

Currently, Transformer has been extensively applied in the field of CD owing to its superior capability in feature extraction and capturing long-term dependencies between images [34], [35]. For example, Li et al. [36] utilized a 2-D self-attention module to extract local spatial–spectral features and improve feature representation. Wang et al. [37] proposed a dual-branch Transformer autoencoder for domain alignment and feature sharing to address limited sample issues. Zhang et al. [38] employed the Transformer-based correlation representation module to explore spectral dependencies. Wang et al. [39] concatenated the bitemporal HSIs to extract temporal information. Zhang et al. [40] proposed a Siamese U-shaped framework, utilizing Swin Transformer blocks to process bitemporal images in parallel. Li et al. [41] proposed a novel hybrid Transformer model and enhanced it with a difference enhancement module to capture change information. These methods utilized Transformer to extract long-term features, and then fused features to obtain the CD results. There are also many methods to extract interactive features. Considering the characteristics of bitemporal HSIs, Song et al. [42] proposed a cross-temporal attention mechanism, which used the spectral attention of one HSI as the temporal attention map of another HSI. Wang et al. [43] proposed a parents-temporal attention module and a brother-spatial attention module, and adopted cross-spatial attention to exploit correlations between bitemporal images. Ding et al. [44] introduced the pixel embedding module and concated the bitemporal HSIs to exploit the correlation and interaction information. Dong et al. [45] proposed a hierarchical multihead self-cross-hybrid attention, which used a cross-correlation matrix to obtain abundant features. Fang et al. [46] proposed an aggregation distribution module to exchange bitemporal features. Although these methods utilize the Transformer model to obtain cross-information and achieve better results, they may overlook the significance of difference information between bitemporal images.

From the above research works, we can see that traditional CNNs often struggle to extract effective features due to their limited local receptive fields, which may be difficult to learn the long-range dependencies. The Transformer-based methods may not fully exploit difference information for CD. To solve these problems, we propose a differential information-enhanced feature exchange network (DIEFEN) for hyperspectral CD. The proposed DIEFEN mainly consists of a channel-spatial interaction (CSI) module, a feature focus and long-range attention (FFLA) module, and an enhanced differential multihead attention (EDMA) module. The CSI integrates channel exchange and spatial exchange mechanisms to align bitemporal features and to diminish the influence of noise. The FFLA consists of a feature compression layer (FCL), a large kernel attention (LKA), and a lightweight layer (LWL), which facilitates the extraction of discriminative features while simultaneously reducing computational complexity. The EDMA module is constructed to guide the original HSIs using difference information, which can effectively suppress unchanged pixels while simultaneously highlight changed pixels. The main contributions of the article are as follows.

1) A novel end-to-end network named DIEFEN is proposed for HSI CD, integrating EDMA, CSI, and FFLA, with the aim of improving the utilization of difference information, extracting local and global features, and reducing sensitivity to noise.
2) The EDMA is proposed to use difference features to enhance original HSI features. It can effectively capture the location and features' changes, and finally improves CD accuracy.
3) The CSI is proposed to align the similarity between the features of the bitemporal HSIs, which can reduce noise through feature interaction between the bitemporal HSIs.

The article is structured as follows: Section II illustrates the proposed method in detail. The experimental results and analysis are provided in Section III. Finally, Section IV concludes the article.

II. Methodology

The framework of the proposed DIEFEN is shown in Fig. 1. First, CSI is performed to facilitate the alignment of features in channel and spatial dimensions. Then, FFLA extracts local features and long-range dependence features. After that, we feed the extracted multiscale features into the EDMA module for feature interaction, using the difference information as a guide to aggregate the features of the bitemporal images and thus to highlight the change pixels. The multiscale difference features extracted by the EDMA modules are then fused and input into the classifier to obtain a binary CD map. The details of DIEFEN are introduced as follows.

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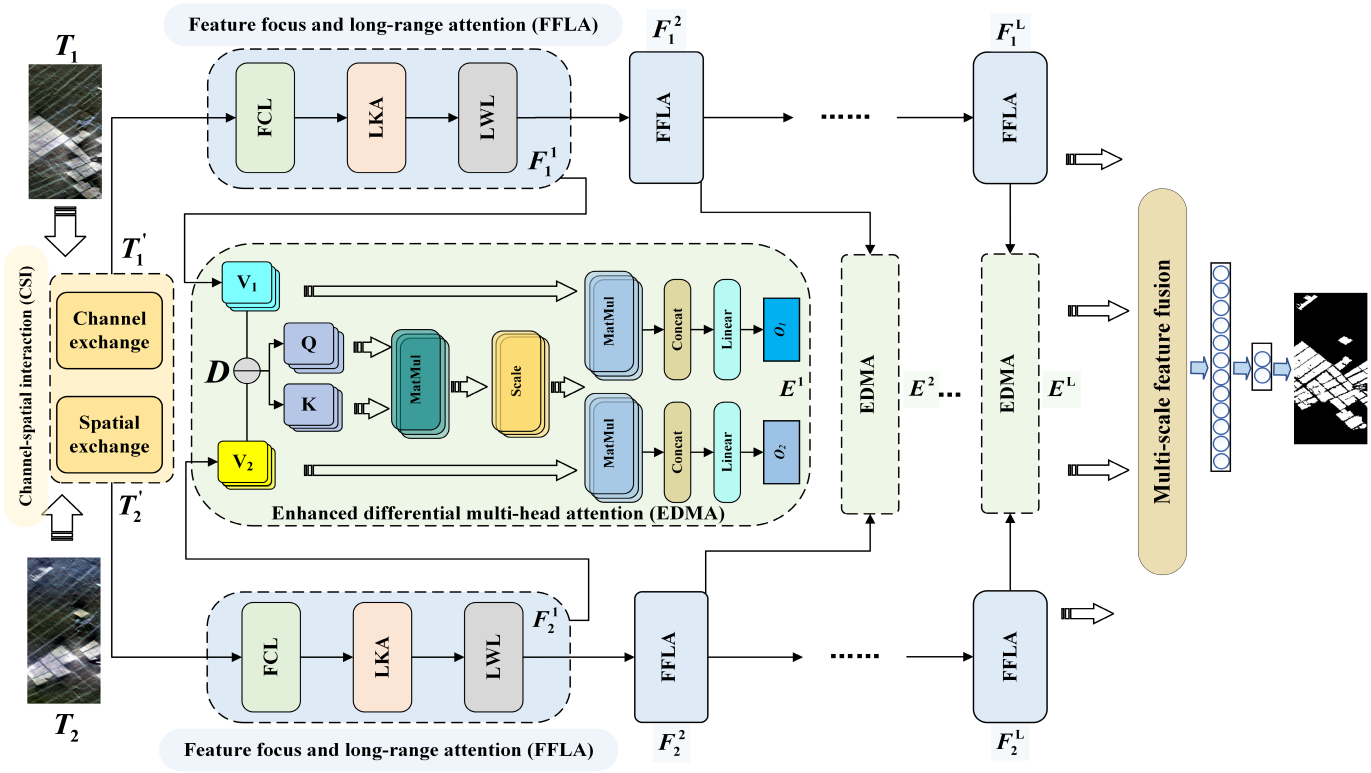


Fig. 1. Framework of the proposed DIEFEN. DIEFEN consists of CSI, FFLA, and EDMA modules. The CSI is used to interact image features. The FFLA is used to capture long-range dependence relation, and EDMA is used to enhance the change information.

A. CSI Module

To alleviate the effect of noise in detecting change pixels between bitemporal HSIs, the CSI is constructed to interact feature along both the spatial and channel dimensions. The CSI module is shown in Fig. 2. Given bitemporal images $\boldsymbol{T}_1 \in \mathbb{R}^{H\times W\times C}$ and $\boldsymbol{T}_2 \in \mathbb{R}^{H\times W\times C}$, the CSI can be represented as

$$\boldsymbol{T}_1'(h,w,c) = \begin{cases} \boldsymbol{T}_1(h,w,c), & \text{if } \boldsymbol{M}(h,w,c)=0 \\ \boldsymbol{T}_2(h,w,c), & \text{if } \boldsymbol{M}(h,w,c)=1 \end{cases} \tag{1}$$

$$\boldsymbol{T}_2'(h,w,c) = \begin{cases} \boldsymbol{T}_2(h,w,c), & \text{if } \boldsymbol{M}(h,w,c)=0 \\ \boldsymbol{T}_1(h,w,c), & \text{if } \boldsymbol{M}(h,w,c)=1 \end{cases} \tag{2}$$

where h, w, and c are the height, width, and channel, respectively. $\boldsymbol{M}$ refers to the exchange mask consisting of 1 and 0, indicating exchange and nonexchange parts. Specifically, we first set an exchange rate p. For the spatial interaction, we randomly select pH rows and pW columns in $\boldsymbol{T}_1$ to exchange with $\boldsymbol{T}_2$. For the channel interaction, we randomly select pC channels in $\boldsymbol{T}_1$ to exchange with $\boldsymbol{T}_2$. After passing through the CSI module, the features of the two images are represented as $\boldsymbol{T}_1'$ and $\boldsymbol{T}_2'$.

By exchanging the spatial and channel features, we can achieve feature alignment along the channel and spatial dimensions, which can effectively suppress unchanged spatial position and shooting angle and enhance the change detector's capacity to identify and capture real change information.

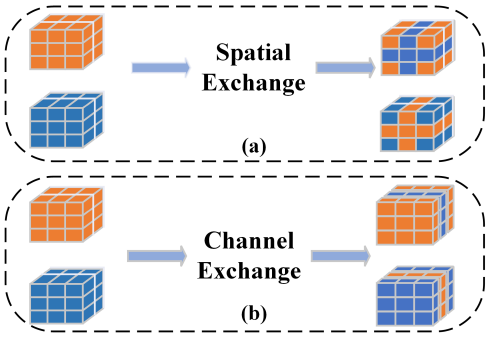


Fig. 2. Schematic of CSI. (a) Spatial exchange, which randomly interacts whole rows and columns of two images. (b) Channel exchange, which randomly interacts the channel dimensions of two images.

B. Feature FFLA Module

To capture long-range dependence relation for subtle CD, we design the multiple FFLAs. Each FFLA includes a FCL, a LKA, and a LWL, as shown in Fig. 3. The FCL employs a 1-D convolutional operation to extract compression features. The LKA is then utilized to extract local contextual information and capture long-range dependence. The LWL can then reduce the computational complexity of the model with depth-wise convolution operation.

Denote the input to each FFLA as intermediate feature $\boldsymbol{F}' \in \mathbb{R}^{\bar{H}\times\bar{W}\times\bar{C}}$, the FCL is

$$\bar{\boldsymbol{K}} = f_{\text{Conv}}^{1\times1}\left(f_{\text{Conv}}^{3\times3}(\boldsymbol{F}')\right) \tag{3}$$

where $f_{\text{Conv}}^{m\times n}$ denotes a conventional convolution with a kernel size of $m \times n$, and the output $\bar{\boldsymbol{K}}$ is of size

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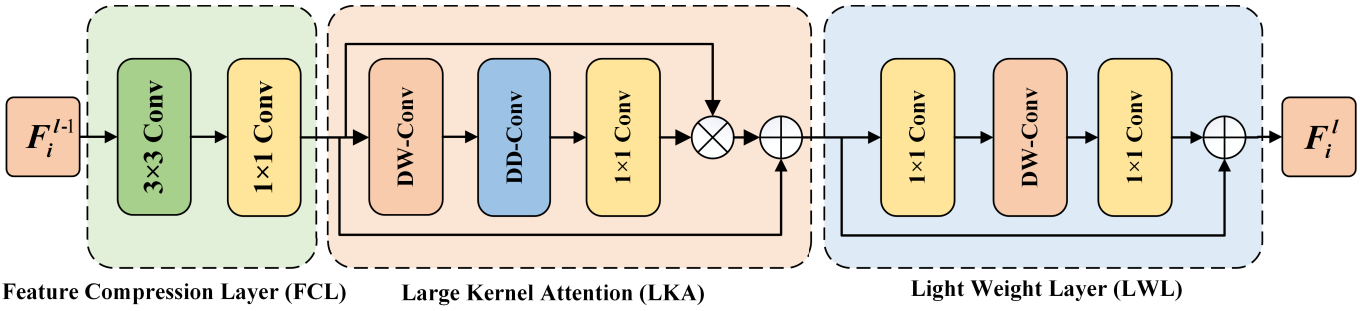


Fig. 3. FFLA structure. FFLA consists of FCL, LKA, and LWL, where FCL is used to reduce the size of feature map, LKA is used to extract local and global information, and LWL is used to reduce the number of parameters.

$(\bar{H}/2) \times (\bar{W}/2) \times 2\bar{C}$. All the convolution layers are followed by the batch normalization and the rectified linear unit (ReLU).

Next, the LKA module is used to capture the local information and the long-range dependencies from HSIs [47]. As shown in Fig. 3, the LKA is composed of a spatial local convolution (depth-wise convolution), a spatial long-range convolution (depth-wise dilation convolution), and a channel convolution (conventional convolution with a kernel size of 1 × 1). Specifically, the LKA module is represented as

$$\hat{\boldsymbol{K}} = f_{\text{Conv}}^{1\times1}\left(f_{\text{DDConv}}^{7\times7}\left(f_{\text{DWConv}}^{5\times5}(\bar{\boldsymbol{K}})\right)\right) \otimes \bar{\boldsymbol{K}} + \bar{\boldsymbol{K}} \tag{4}$$

where $f_{\text{DWConv}}^{m\times n}$ denotes a depth-wise convolution with a kernel size of $m \times n$, $f_{\text{DDConv}}^{m\times n}$ denotes a depth-wise dilation convolution with a kernel size of $m \times n$, and $\otimes$ means element-wise product. Note that, the residual connect is used in the LKA to alleviate the vanishing gradient problem.

Finally, the feature enhance layer of the FFLA is

$$\boldsymbol{F} = f_{\text{Conv}}^{1\times1}\left(f_{\text{DWConv}}^{3\times3}\left(f_{\text{Conv}}^{1\times1}(\hat{\boldsymbol{K}})\right)\right) + \hat{\boldsymbol{K}}. \tag{5}$$

This module addresses the constraints of traditional CNNs related to local receptive fields and is capable of extracting multiple features. The FFLA uses depth-wise convolution to capture local contextual details, applies depth-wise dilated convolution to identify long-range dependencies, and utilizes a 1 × 1 kernel size to achieve channel-wise self-adaptation. By combining several FFLAs, our model effectively identifies large-scale change regions and detects fine-grained changes.

C. EDMA Module

Most existing transformer-based HSI CD methods extract features using a cross-attention strategy. However, these methods do not effectively utilize change information to guide the model's focus on the changed regions. To address this limitation, we designed the EDMA module to highlight change information using differential features. The EDMA module specifically extracts and amplifies differential features between bitemporal images, ensuring that the model's focus is directed toward regions of change. Fig. 4 illustrates the detailed process of differential enhancement in the EDMA module.

Let $\boldsymbol{F}_1 \in \mathbb{R}^{\bar{H}\times\bar{W}\times\bar{C}}$ and $\boldsymbol{F}_2 \in \mathbb{R}^{\bar{H}\times\bar{W}\times\bar{C}}$ denote the features extracted by the FFLA, respectively. The EDMA first reshapes them along the spatial dimension to obtain $\hat{\boldsymbol{F}}_1 \in \mathbb{R}^{\bar{H}\bar{W}\times\bar{C}}$ and $\hat{\boldsymbol{F}}_2 \in \mathbb{R}^{\bar{H}\bar{W}\times\bar{C}}$, and then gets the difference feature $\boldsymbol{D}$ by

$$\boldsymbol{D} = \hat{\boldsymbol{F}}_1 - \hat{\boldsymbol{F}}_2. \tag{6}$$

Next, the EDMA obtains the query and the key from the difference feature $\boldsymbol{D}$, and also the values of bitemporal images from $\hat{\boldsymbol{F}}_1$ and $\hat{\boldsymbol{F}}_2$, respectively. The formula is as follows:

$$\begin{cases} \boldsymbol{Q} = \boldsymbol{D}\boldsymbol{W}_q \\ \boldsymbol{K} = \boldsymbol{D}\boldsymbol{W}_k \\ \boldsymbol{V}_1 = \hat{\boldsymbol{F}}_1\boldsymbol{W}_{v_1} \\ \boldsymbol{V}_2 = \hat{\boldsymbol{F}}_2\boldsymbol{W}_{v_2} \end{cases} \tag{7}$$

where $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_{vi} \in \mathbb{R}^{\bar{C}\times d_k}$ are weight matrices for getting query, key, and values, respectively, and d_k is the channel dimension.

Afterward, the EDMA calculates the attention score value guided by the query and key from difference feature, and then extracts features $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ for bitemporal images. The specific formula is as follows:

$$\begin{cases} \boldsymbol{H}_1 = \text{softmax}\left(\dfrac{\boldsymbol{Q}\times\boldsymbol{K}^{\text{T}}}{\sqrt{d_k}}\right)\boldsymbol{V}_1 \\ \boldsymbol{H}_2 = \text{softmax}\left(\dfrac{\boldsymbol{Q}\times\boldsymbol{K}^{\text{T}}}{\sqrt{d_k}}\right)\boldsymbol{V}_2. \end{cases} \tag{8}$$

In the EDMA, the multihead attention mechanism is utilized. Specifically, assume there are K independent attention mechanisms, we can obtain K features $\boldsymbol{H}_1^k (k = 1, 2, \ldots, K)$ for $\boldsymbol{T}_1$, and K features $\boldsymbol{H}_2^k (k = 1, 2, \ldots, K)$ for $\boldsymbol{T}_2$. Then, the obtained features are concatenated and then they are fed into a fully connected (FC) layer, resulting in the following output:

$$\boldsymbol{O}_i = \text{Concat}\left(\boldsymbol{H}_i^1, \boldsymbol{H}_i^2, \ldots, \boldsymbol{H}_i^K\right)\boldsymbol{W}_{O_i} \tag{9}$$

where $i = 1, 2$, and Concat$(\cdot)$ denotes the concatenation operation.

Finally, the EDMA concatenates $\boldsymbol{D}$, $\boldsymbol{O}_1$, and $\boldsymbol{O}_2$, generating the output $\boldsymbol{E}$ as

$$\boldsymbol{E} = \text{Concat}(\boldsymbol{D}, \boldsymbol{O}_1, \boldsymbol{O}_2). \tag{10}$$

After the EDMA module, we can enhance the original image by highlighting differential features, effectively emphasizing change information and suppressing unchanged information.

D. DIEFEN-Based CD

Given bitemporal images $\boldsymbol{T}_1 \in \mathbb{R}^{H\times W\times C}$ and $\boldsymbol{T}_2 \in \mathbb{R}^{H\times W\times C}$, the DIEFEN first feeds them into the

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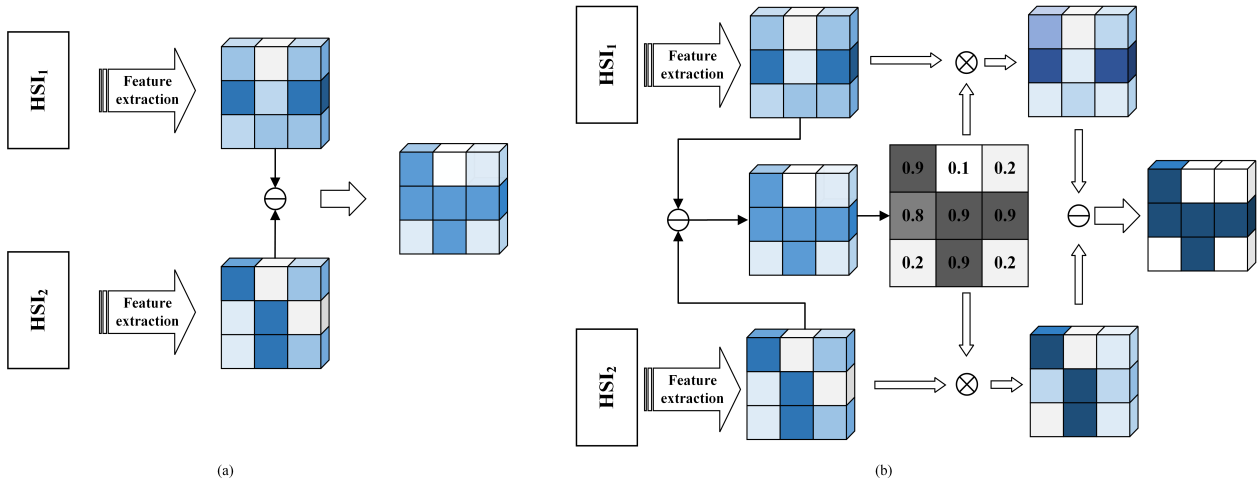


Fig. 4. Schematic of differential enhancement. The extracted features are shown in different colors. (a) Direct subtraction of bitemporal images. (b) Differential enhancement that employs enhanced attention mechanisms prior to subtraction to amplify the change features while diminishing unchanged features.

CSI to align features along spatial and spectral dimensions, which can be formulated as

$$\left(\boldsymbol{T}_1', \boldsymbol{T}_2'\right) = \mathrm{CSI}(\boldsymbol{T}_1, \boldsymbol{T}_2). \tag{11}$$

Then, the obtained $\boldsymbol{T}_1'$ and $\boldsymbol{T}_2'$ are input into two FFLA-based branches to extract features. We utilize L FFLA modules in each branch, and also use the downsampling operation between any two FFLA modules to mine multiscale features. The formula can be represented as

$$\begin{cases} \boldsymbol{F}_1^l = \mathrm{DS}\left(\mathrm{FFLA}^l\left(\boldsymbol{F}_1^{l-1}\right)\right) \\ \boldsymbol{F}_2^l = \mathrm{DS}\left(\mathrm{FFLA}^l\left(\boldsymbol{F}_2^{l-1}\right)\right) \end{cases} \tag{12}$$

where $l = 1, 2, \ldots, L$, and $\mathrm{DS}(\cdot)$ denotes the downsampling operation. $\boldsymbol{F}_i^l \in \mathbb{R}^{(H/(l+1))\times(W/(l+1))\times d(l+1)}$ and $\boldsymbol{F}_i^{l-1} \in \mathbb{R}^{(H/l)\times(W/l)\times dl}$ $(i = 1, 2)$ represent the outputs and inputs of lth FFLA module, respectively, and here d denotes the feature dimension. Besides, we set $\boldsymbol{F}_1^0 = f_{\mathrm{Conv}}^{1\times1}(f_{\mathrm{Conv}}^{7\times7}(\boldsymbol{T}_1')) \in \mathbb{R}^{H\times W\times d}$ and $\boldsymbol{F}_2^0 = f_{\mathrm{Conv}}^{1\times1}(f_{\mathrm{Conv}}^{7\times7}(\boldsymbol{T}_2')) \in \mathbb{R}^{H\times W\times d}$.

Next, the output pairs of all FFLA modules, namely $\{\boldsymbol{F}_1^l, \boldsymbol{F}_2^l\}_{l=1}^L$, are fed into the corresponding EDMA modules as

$$\boldsymbol{E}^l = \mathrm{EDMA}^l\left(\boldsymbol{F}_1^l, \boldsymbol{F}_2^l\right) \tag{13}$$

where $l = 1, 2, \ldots, L$.

Finally, the extracted multiscale features by multiple EDMA modules are concatenated to obtain the final difference features

$$\boldsymbol{P} = \mathrm{Concat}\left(\mathrm{U}\left(\mathrm{GAP}\left(\boldsymbol{E}^1\right)\right), \mathrm{U}\left(\mathrm{GAP}\left(\boldsymbol{E}^2\right)\right), \ldots, \mathrm{U}\left(\mathrm{GAP}\left(\boldsymbol{E}^L\right)\right)\right) \tag{14}$$

where $\mathrm{GAP}(\cdot)$ denotes the global average pooling operation. $\mathrm{U}(\cdot)$ denotes feature upsampling to 512-D. The features $\boldsymbol{P}$ are then fed into a softmax layer to obtain the CD output $\hat{y}$

$$\hat{y} = \mathrm{softmax}(\boldsymbol{P}\boldsymbol{W}) \tag{15}$$

where $\boldsymbol{W}$ denotes the weight of the softmax layer.

In HSI CD task, each pixel of the bitemporal HSIs is classified into two categories: changed class and unchanged class. The cross-entropy loss function is used for network optimization as

$$\mathrm{Loss} = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log \hat{y}_i + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right) \tag{16}$$

where n denotes the number of labeled samples, and y_i and $\hat{y}_i$ denote the real and predicted value of the ith sample, respectively.

TABLE I
DETAILED INFORMATION OF THREE PUBLIC DATASETS

	Farmland	River	USA
Size	450×140	463×241	307×241
Channels	155	198	154
Changed	18277	9698	16676
Unchanged	44723	101885	57311
Data of T1	May 3, 2006	May 3,2013	May 1, 2004
Data of T2	Apr 23, 2007	Dec 31, 2013	May 8, 2007

III. EXPERIMENTS

In this section, experiments are conducted to validate the effectiveness of the proposed DIEFEN.

A. Experimental Setup

1) Dataset: Three widely used datasets are used, namely Farmland, River, and USA. The detailed information of the three datasets is given in Table I.

Farmland dataset was collected by the Earth observing (EO)-1 Hyperion sensor over a wetland agricultural area in Yancheng City, Jiangsu, China, on May 3, 2006, and April 23, 2007. After removing noise and water-absorption bands, Farmland dataset contains 420 × 140 pixels and 155 spectral bands. The main change is about the cultivation changes. Fig. 5 shows the Farmland dataset.

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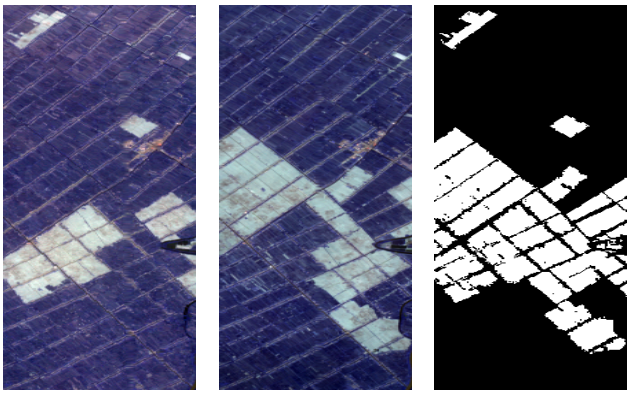


Fig. 5. Farmland dataset. (a) Imagery acquired on May 3, 2006. (b) Imagery acquired on April 23, 2007. (c) Ground truth.

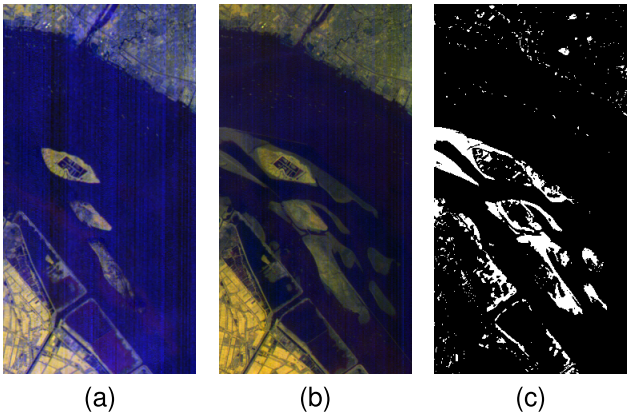


Fig. 6. River dataset. (a) Imagery acquired in May 2013. (b) Imagery acquired in December 2013. (c) Ground truth.

River dataset was obtained by the EO-1 Hyperion over a river area in Jiangsu, China, on May 3, 2013, and December 31, 2013. After removing noise, the image with 463 × 241 pixels and 198 bands remained. The primary change type is about the transfer and disappearance of water quality areas. Fig. 6 shows the River dataset.

USA dataset contains two HSIs captured on May 1, 2004, and May 8, 2007, respectively. USA dataset is collected from an irrigated agricultural field in Hermiston City in Umatilla County, Oregon, USA by Hyperion sensor. The USA image contains 307 × 241 pixels and 154 bands ranging from 0.4 to 2.5 um. The main changes on the datasets are mainly the sizeable changes in cultivated land and relatively fine riverbank line changes. Fig. 7 shows the USA dataset.

2) Evaluation Metrics: Five commonly used metrics are used to evaluate the performance, including overall accuracy (OA), Kappa Coefficient (κ), Precision (P), Recall (R), and $F1$-score ($F1$). These metrics are defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FN} + \mathrm{FP}} \tag{17}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{18}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{19}$$

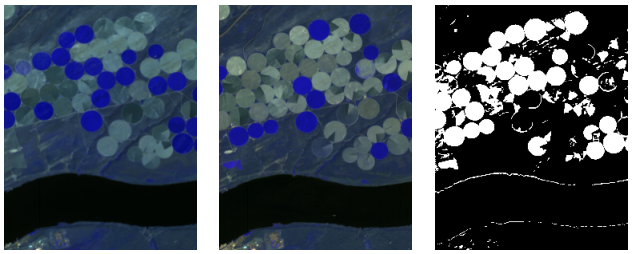


Fig. 7. USA dataset. (a) Imagery acquired in 2004. (b) Imagery acquired in 2007. (c) Ground truth.

$$F1 = \frac{2P \times R}{P + R} \tag{20}$$

$$\kappa = \frac{\mathrm{OA} - p_e}{1 - p_e} \tag{21}$$

$$p_e = \frac{(\mathrm{TP} \cdot \mathrm{FN}) + (\mathrm{TP} \cdot \mathrm{FP}) + (\mathrm{TN} \cdot \mathrm{FN}) + (\mathrm{TN} \cdot \mathrm{FP})}{N^2} \tag{22}$$

where true positive (TP) denotes the number of pixels that are correctly classified in the change regions, true negative (TN) represents the number of pixels that are correctly classified in the unchanged areas, false positive (FP) denotes the number of misclassified pixels in the change regions, and false negative (FN) means the number of misclassified pixels in the unchanged areas. Greater values of the above five metrics indicate superior performance.

3) Comparative Models: The DIEFEN is compared with several state-of-the-art HSI CD methods including CVA [48], PCA-CVA [12], MAD [49], SVM [50], BIT [51], ConvLSTM [52], CSANet [42], and SSTFormer [39]. CVA is a method that identifies changes by calculating the Euclidean distance between pixels. PCA-CVA combines the PCA for dimensionality reduction with the CVA for CD. MAD employs the PCA and canonical correlation analysis for data transformation and difference analysis. SVM processes image data by normalizing and superimposing spectral data, and classifying it using an radial basis function kernel function. BIT uses a transformer network to map input image features. ConvLSTM combines CNN with long short-term memory networks to learn multiscale features. CSANet employs cross-temporal attention to enhance feature extraction. SSTFormer concatenates bitemporal images to extract temporal features.

We randomly select 1% labeled pixels for training and the rest pixels are used for testing. For deep learning-based methods, the SGD optimizer with a weight decay of 5e-3 is used, and the number of epochs is set to 100. For the BIT, ConvLSTM, and CSANet, the input size is 9 × 9. For the SSTFormer, the input size is 5 × 5. The input patch size of our DIEFEN is 5×5, and the number of FFLA is $L = 4$. The experiments are repeated ten times and the averaged results and standard results are reported.

B. Experimental Results

1) Farmland Dataset: Table II and Fig. 8 show the results of all methods on the Farmland dataset. The analysis reveals that supervised deep learning approaches have significant

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TABLE II
CD RESULTS ON THE FARMLAND DATASET WITH 1% TRAINING SAMPLES

Method	OA%	$\kappa\times100$	F1%	P%	R%
CVA	95.74	89.83	92.87	90.36	95.51
PCA-CVA	95.78	89.91	92.91	90.54	95.41
MAD	96.18	90.61	93.26	**95.81**	90.84
SVM	96.24 ± 0.13	90.93 ± 0.28	93.59 ± 0.18	92.40 ± 0.86	94.83 ± 0.64
BIT	95.77 ± 0.33	89.88 ± 0.71	92.89 ± 0.46	90.70 ± 1.71	95.34 ± 1.26
ConvLSTM	95.45 ± 0.52	89.10 ± 1.18	92.09 ± 0.97	92.89 ± 1.26	91.34 ± 2.18
CSANet	96.48 ± 0.20	91.47 ± 0.45	93.95 ± 0.31	93.79 ± 1.28	94.13 ± 1.06
SSTFormer	96.18 ± 0.41	90.83 ± 0.96	93.52 ± 0.68	92.39 ± 0.91	94.68 ± 0.92
DIEFEN	**96.86 ± 0.24**	**92.48 ± 0.57**	**94.70 ± 0.39**	92.77 ± 0.91	**96.74 ± 0.66**

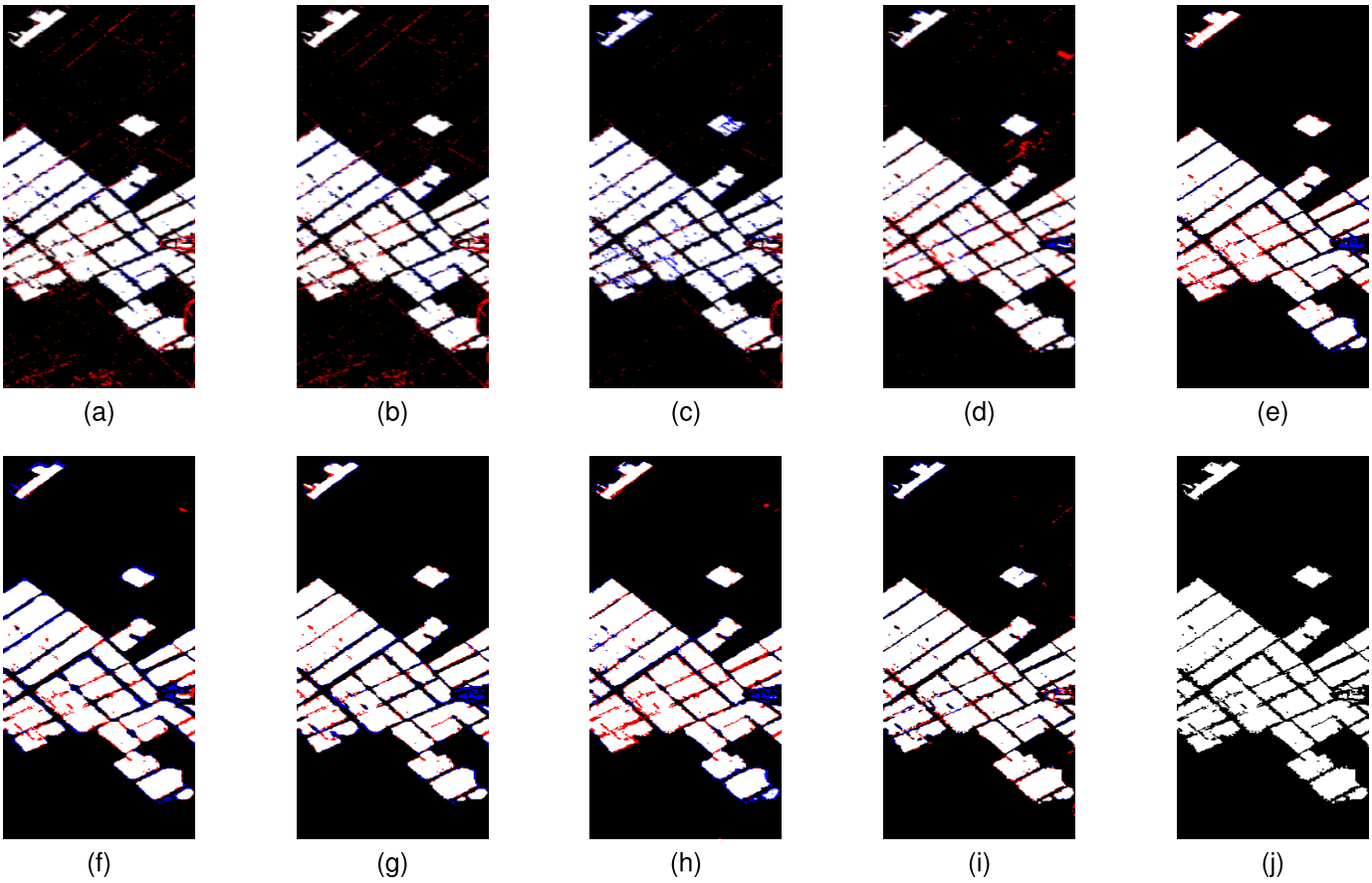


Fig. 8. CD map of all methods on the farmland dataset. (a) CVA. (b) PCA-CVA. (c) MAD. (d) SVM. (e) BIT. (f) ConvLSTM. (g) CSANet. (h) SSTFormer. (i) DIEFEN. (j) GT. The FN and FP pixels are denoted in red and blue colors, respectively.

advantages over traditional unsupervised algorithms. Although classical methods, such as CVA and PCA-CVA, achieve high recall values, they suffer from low precision, resulting in a higher probability of FPs where unchanged pixels are incorrectly classified as changed ones. Among the deep learning algorithms, ConvLSTM has a high P value but a relatively lower R value than other algorithms, indicating that the algorithm may incorrectly predict more changed pixels as unchanged ones. Compared to other algorithms, the proposed DIEFEN has higher OA, κ, $F1$, and R values. As DIEFEN uses the difference information of bitemporal images to highlight change information and suppress unchanged information, it can generate relatively larger TP and smaller FN, and can effectively detect changed and unchanged pixels.

Fig. 8 shows that our proposed DIEFEN algorithm has the lowest false detection rate and produces the best visual results. In contrast, the classical CVA and PCA-CVA algorithms exhibit a high degree of false detection, whereby unchanged regions are identified as changed and pixels located at the

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TABLE III
CD RESULTS ON THE RIVER DATASET WITH 1% TRAINING SAMPLES

Method	OA%	$\kappa\times100$	F1%	P%	R%
CVA	95.52	66.08	68.36	88.42	55.72
PCA-CVA	95.49	65.61	67.88	89.06	54.83
MAD	92.87	52.38	56.24	60.23	52.75
SVM	94.88 ± 0.91	60.47 ± 1.72	59.73 ± 1.27	**92.14 ± 1.53**	46.19 ± 1.36
BIT	94.92 ± 0.13	62.76 ± 2.10	65.39 ± 2.09	80.29 ± 1.98	55.33 ± 3.89
ConvLSTM	94.86 ± 0.19	63.78 ± 2.10	66.50 ± 2.13	76.98 ± 4.57	59.02 ± 5.36
CSANet	95.16 ± 0.28	66.05 ± 2.51	68.63 ± 2.40	78.49 ± 2.23	61.10 ± 3.87
SSTFormer	94.99 ± 0.66	65.83 ± 6.48	68.50 ± 6.23	75.01 ± 3.56	63.41 ± 8.65
DIEFEN	**96.06 ± 0.14**	**74.27 ± 0.97**	**76.41 ± 0.91**	79.67 ± 1.74	**73.49 ± 2.22**

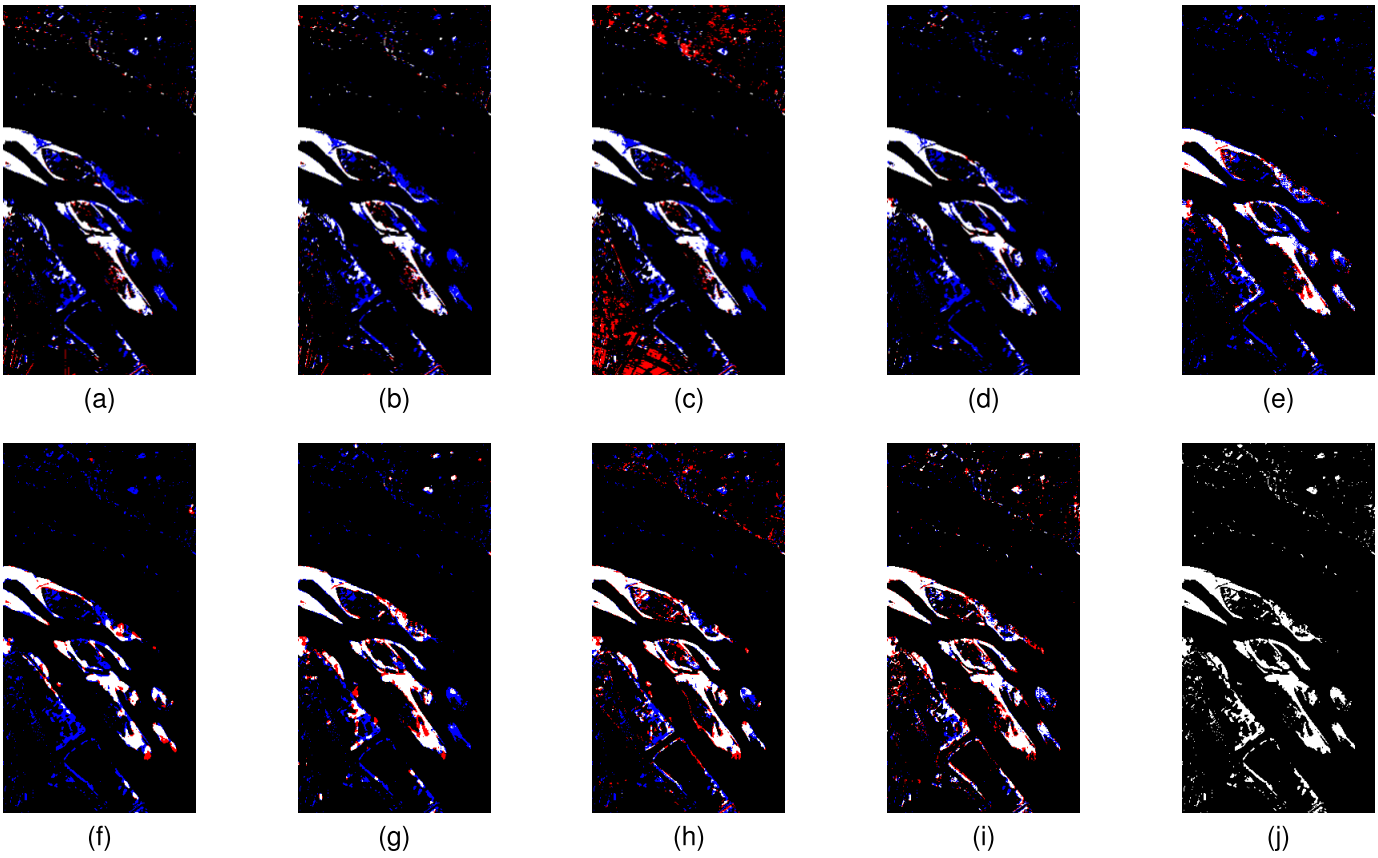


Fig. 9. CD map of all methods on the river dataset. (a) CVA. (b) PCA-CVA. (c) MAD. (d) SVM. (e) BIT. (f) ConvLSTM. (g) CSANet. (h) SSTFormer. (i) DIEFEN. (j) GT. The FN and FP pixels are denoted in red and blue colors, respectively.

edges of changed areas and small targets are misclassified. The false detection rate of MAD and SVM algorithms are lower than that of CVA and PCA-CVA, but there are still some apparent false detected pixels. The deep learning algorithms, BIT, ConvLSTM, CSANet, and SSTFormer, perform well compared to traditional methods. However, they may not be able to detect subtle information and fine boundary details. Compared to other deep learning methods, the proposed DIEFEN has fewer misclassified points and provides better CD details.

2) River Dataset: Table III presents the results of various algorithms on the River dataset. Traditional methods have high OA values but low R values. This is especially noticeable in the MAD algorithm. In the case of limited labeled pixels, it is difficult to distinguish changing features when the changed and unchanged features are highly similar. Consequently, the traditional methods predict most of the changed pixels as unchanged ones. Deep learning methods identify changed pixels more accurately than most traditional algorithms. Our proposed DIEFEN outperforms other models with the highest

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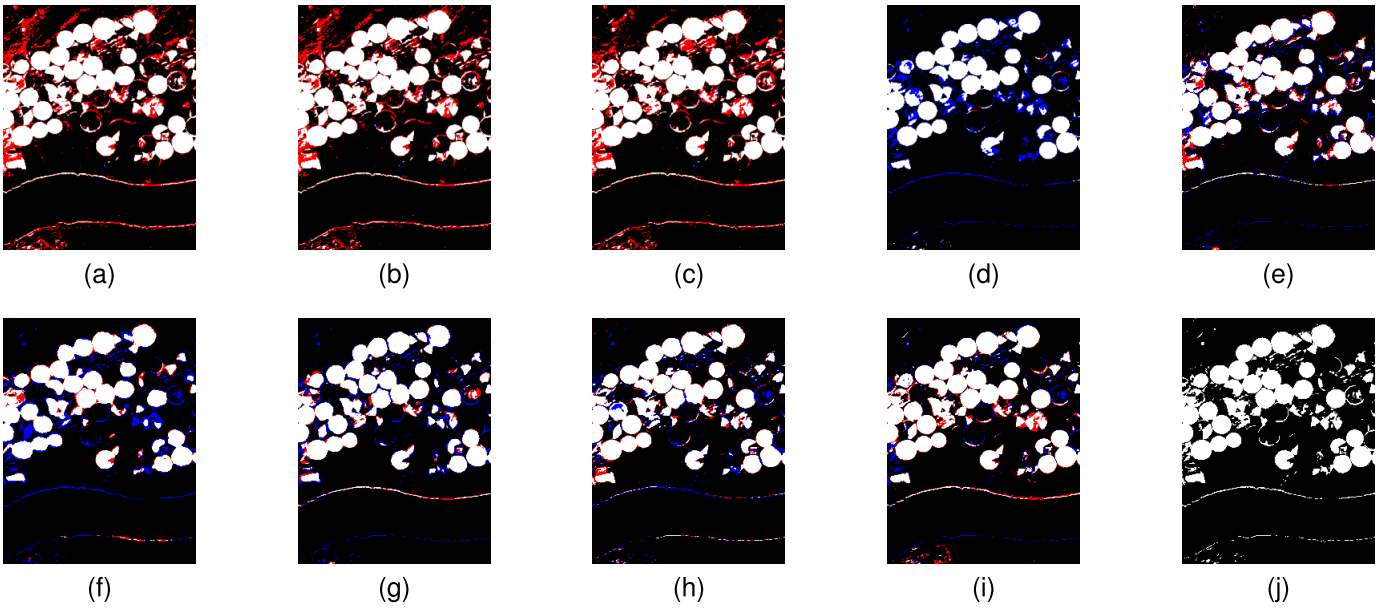


Fig. 10. CD maps of all methods on the USA dataset. (a) CVA. (b) PCA-CVA. (c) MAD. (d) SVM. (e) BIT. (f) ConvLSTM. (g) CSANet. (h) SSTFormer. (i) DIEFEN. (j) GT. The FN and FP pixels are denoted in red and blue colors, respectively.

TABLE IV
CD RESULTS ON THE USA DATASET WITH 1% TRAINING SAMPLES

Method	OA%	$\kappa\times100$	F1%	P%	R%
CVA	90.76	76.87	82.95	71.02	99.72
PCA-CVA	91.23	77.89	83.68	72.09	99.68
MAD	91.70	78.95	84.42	73.18	**99.75**
SVM	92.94 ± 2.41	78.25 ± 8.04	82.58 ± 6.69	91.88 ± 5.96	75.96 ± 10.59
BIT	94.23 ± 0.64	83.41 ± 1.80	87.12 ± 1.38	87.71 ± 2.10	86.58 ± 1.43
ConvLSTM	93.46 ± 0.32	80.43 ± 1.08	84.56 ± 0.93	90.42 ± 2.29	79.52 ± 2.75
CSANet	94.38 ± 0.33	83.30 ± 1.02	86.86 ± 0.82	91.80 ± 1.52	82.46 ± 1.67
SSTFormer	95.13 ± 0.15	86.25 ± 0.33	89.80 ± 0.17	**91.08 ± 2.04**	89.09 ± 2.41
DIEFEN	**95.58 ± 0.29**	**87.53 ± 0.83**	**90.41 ± 0.66**	90.22 ± 2.61	92.31 ± 2.83

R value. This demonstrates that the designed differential enhancement module can accurately identify changed pixels.

Fig. 9 illustrates that traditional CD algorithms have a higher rate of false detections and missing detections. This is evident in Fig. 9(c), where the MAD predicts many unchanged pixels as changed pixels (red regions). Deep learning algorithms can detect unchanged pixels well, but miss more changed pixels. ConvLSTM, BIT, and SSTFormer fail to detect changed pixels in the upper side region of the image. Although the CSANet detects some of the changed pixels in the upper side region of the image, it still detects fewer pixels than our proposed DIEFEN. Our DIEFEN has fewer misclassified pixels (red and blue regions) and provides better CD details compared to other algorithms.

3) USA Dataset: Table IV shows the results of different methods on the USA dataset. Traditional algorithms accurately detect almost all changed pixels, but tend to make errors in detecting unchanged pixels, resulting in lower OA and Kappa values. The SVM has a higher *P* and detects almost all unchanged pixels, but detects changed pixels poorly. The USA dataset contains scattered changed regions without large connected areas. These features can cause misclassification, potentially weakening the deep learning network's detection ability compared to traditional algorithms. The DIEFEN achieves higher accuracy than CSANet, ConvLSTM, and SSTFormer. A possible reason is that the DIEFEN can better capture and obtain change features of the bitemporal HSIs. In general, the DIEFEN shows overall better results than other methods.

Fig. 10 shows the visual observation results on the USA dataset. The CVA, PCA-CVA, and MAD predict unchanged pixels as changed pixels. While SVM misclassifies many pixels in the change regions due to insufficient training samples, deep learning methods, such as, BIT, ConvLSTM, and CSANet, can improve accuracy compared to traditional methods, they also show poor performance in boundary regions. Our DIEFEN generates the best visual results with the smallest FN and FP pixels.

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TABLE V
ABLATION EXPERIMENTS ON THE THREE DATASETS

Dataset	FFLA	CSI	EDMA	OA%	$\kappa\times100$	F1%	P%	R%
Farmalnd	✓	×	×	96.08 ± 0.42	90.54 ± 1.06	93.32 ± 0.77	92.32 ± 1.15	94.37 ± 1.95
	✓	×	✓	96.16 ± 0.07	90.87 ± 0.18	93.61 ± 0.13	90.44 ± 0.06	**97.02 ± 0.35**
	✓	✓	×	96.27 ± 0.34	90.76 ± 0.99	93.34 ± 0.79	**94.15 ± 1.37**	94.49 ± 1.87
	✓	✓	✓	**96.86 ± 0.24**	**92.48 ± 0.57**	**94.70 ± 0.39**	92.77 ± 0.91	96.74 ± 0.66
River	✓	×	×	95.21 ± 0.05	64.37 ± 1.24	66.83 ± 1.31	77.89 ± 2.65	55.74 ± 2.27
	✓	×	✓	95.87 ± 0.20	72.60 ± 1.55	74.84 ± 1.44	79.62 ± 1.26	70.59 ± 1.42
	✓	✓	×	95.14 ± 0.13	66.28 ± 1.55	68.88 ± 1.51	77.69 ± 1.45	61.93 ± 1.97
	✓	✓	✓	**96.06 ± 0.14**	**74.27 ± 0.97**	**76.41 ± 0.91**	**79.67 ± 1.74**	**73.49 ± 2.22**
USA	✓	×	×	94.56 ± 0.04	84.22 ± 0.44	87.71 ± 0.50	87.32 ± 0.19	88.49 ± 0.32
	✓	✓	×	94.54 ± 0.43	84.54 ± 1.01	88.04 ± 0.73	86.80 ± 1.73	89.42 ± 1.14
	✓	×	✓	94.81 ± 0.15	83.80 ± 0.34	87.54 ± 0.24	87.76 ± 1.35	89.78 ± 1.43
	✓	✓	✓	**95.58 ± 0.29**	**87.53 ± 0.83**	**90.41 ± 0.66**	**90.22 ± 2.61**	**92.31 ± 2.83**

4) Ablation Study: In this section, ablation experiments are conducted on three datasets to demonstrate the effectiveness of different modules, namely the FFLA, CSI, and EDMA. To this end, we design three variants. The first variant uses only the FFLA for feature extraction, followed by direct difference calculation, feature fusion, and classification to obtain the CD result. The second variant incorporates the EDMA module into the first variant to highlight image regions with differences. The third variant integrates the CSI module into the first variant for feature alignment between bitemporal images.

Table V shows that the accuracy improves as the number of modules increases. Our proposed model achieves the highest overall accuracy, demonstrating the effectiveness of each module in improving accuracy. For the Farmland dataset, the addition of only the EDMA module results in a significant increase in R, indicating that EDMA can detect more detailed information about changes. Similarly, adding only the CSI module improves the model's OA, κ, and $F1$ values. This indicates that the CSI module can reduce noise, ultimately improving the model's accuracy in recognizing and capturing change information. On the River dataset, adding the EDMA module alone also improves the recall dramatically. By further adding the CSI module, all five indices are improved. Similar results also can be found on the USA dataset.

C. Discussion

1) Discussion on the Number of FFLA Layers: Fig. 11 shows the effect of the number of FFLA layer L, where L changes from 2 to 5. For small L, the model with a shallow architecture fails to capture the complex spatial and spectral relationships in HSIs and generates poor results. As the layer increases, the model performance generally improves. It can be seen that our model shows the best results in the case of $L = 4$.

2) Discussion on the Mask Strategy in the CSI Module: In the CSI module, we consider different mask ratio p, i.e., $p = 0.1$, $p = 0.3$, $p = 0.5$, $p = 0.7$, and $p = 0.9$, and show the corresponding results in Fig. 12. It can be seen that our model shows the best results in the case of $p = 0.5$. For small mask ratio, the module cannot extract sufficient interactive features. For large mask ratio, with too many interactive information, the module may be meaningless.

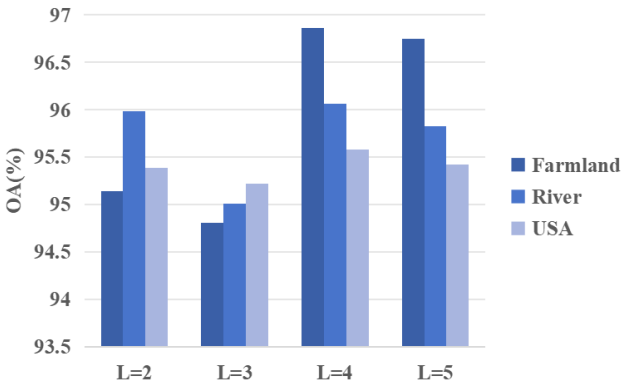


Fig. 11. Effect of the number of FFLA layers L.

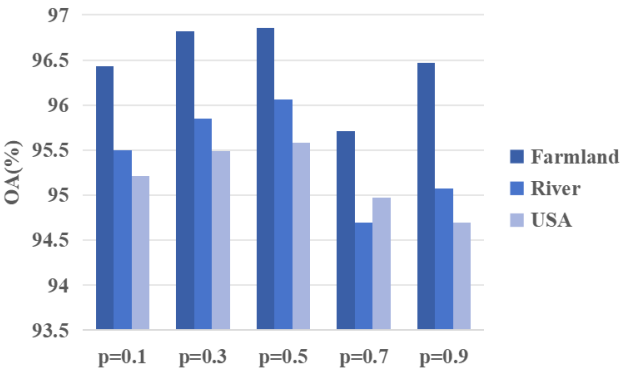


Fig. 12. Effect of the mask strategy in the CSI module.

3) Discussion on the Total Parameters and Flops: The total parameters and FLOPs of different methods are shown

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TABLE VI
TOTAL PARAMETERS AND FLOPS OF DIFFERENT METHODS ON THE FARMLAND DATASET

Method	Parameters	FLOPs
BIT	12.52M	6.98G
ConvLSTM	0.24M	13.63G
CSANet	40.42M	14.59G
SSTFormer	2.50M	8.11G
DIEFEN	8.08M	7.20G

in Table VI. Although ConvLSTM has the fewest parameters, it exhibits the highest FLOPs, indicating it performs more operations per parameter. CSANet has the highest number of parameters and FLOPs. Compared with other methods, our proposed DIEFEN balances parameters and FLOPs.

IV. CONCLUSION

This article proposes an end-to-end DIEFEN for detecting changes in bitemporal HSIs. The DIEFEN mainly includes three modules, i.e., FFLA, CSI, and EDMA. The proposed FFLA combines the FCL, LKA, and LWL to extract long-range features from the HSIs. By randomly exchanging channel and spatial dimensions, the designed CSI can reduce noise and align features between bitemporal HSIs. The EDMA module effectively highlights changes pixels while suppressing unchanged pixels based on the bitemporal images' difference information. Experimental results on three HSI CD datasets demonstrate that our proposed method yields more accurate results compared to other comparative methods especially for small sample size.

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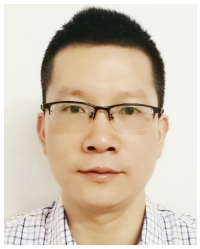
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